THE IMPACT OF VOCAL MUSIC IN PUBLIC SOUND ENVIRONMENTS

Group KF1-A

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Abstract

This project explored how vocals in music affect cognitive performance and stress. An eperimantal setup let perticipants conduct two different tasks while listening to either soft or hard vocals in music. This was done in a whithin group design. NASA-TLX and STAI surveys allowed participants to self record their experience during the experiment. Interviews rounded up each experiment session. The results show that there is no significant difference between the two vocal types and the effect they have on performance.

Background

Previous research has shown that music can influence cognitive performance, mood, and perceived stress levels. Studies on background music have found that factors such as tempo, loudness, and musical complexity can support or interfere with mood and task performance. Vocal music has been mentioned as especially distracting because lyrics can occupy the same cognitive resources needed to read, memorize, and perform mentally demanding tasks.

While research has compared vocal versus instrumental music, fewer studies have examined the impact of different levels or types of vocals and how they affect mood and cognitive abilities. This distinction is important to provide a new perspective to the area of music perception, and highlights the need for studies that explore how varying vocal music influences the everyday working conditions of employees in public spaces.

The aim of this project is to explore how different vocals in music affect anxiety levels and cognitive performance.

Method

Design: Within-subject design
Participants: Adults (21- 31 years old) without hearing or cognitive impairment were randomly divided into two groups.
Setting: Lab simulated convenience store environment.

Equipment: Tasks performed on a laptop with audio delivered via external speakers. Web-based cognitive assessment platforms were used for tasks.

Procedure:

1. **Pre-surveys:** State-Trait Anxiety Inventory (STAI)
2. **Cognitive tasks:** Digital span and symbol search.
3. **Post-surveys:** STAI and NASA Task Load Index
4. **Interview**

Audio Stimuli Generation

Realistic recordings of a convenience store (talking, coffee machines).
Using udio.ai to create music with distinct vocal styles (rough and soft) while keeping tempo and volume constant.

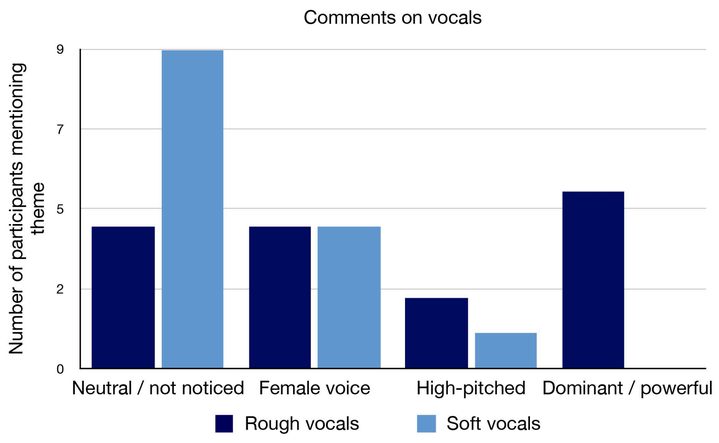
Have a listen

Results

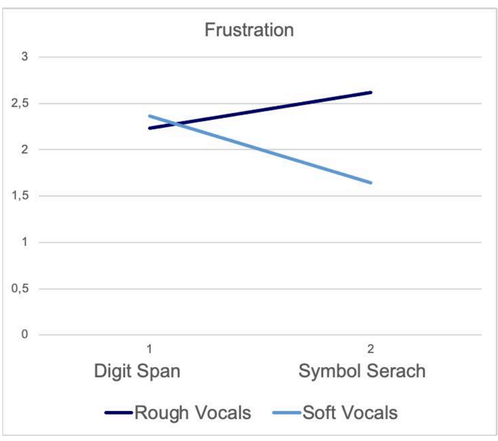
The results from the STAI survey show that participants had significantly higher levels of anxiety after experiments (M=41.1, SD=7.9) compared to before (M=35.8, SD=6.6), $F(1, 33.93) = 23.94$, $p < .001$.

Task performance **differed slightly** between the soft and hard vocal conditions, with a small advantage for soft vocals. However, these differences were **not statistically significant** and varied across participants.

Graph 1. Interview data



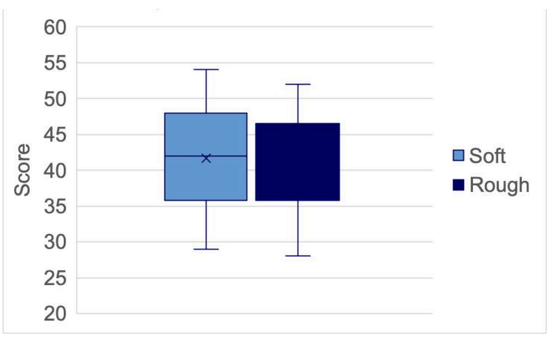
Graph 2. NASA-TLX Survey Result on frustration



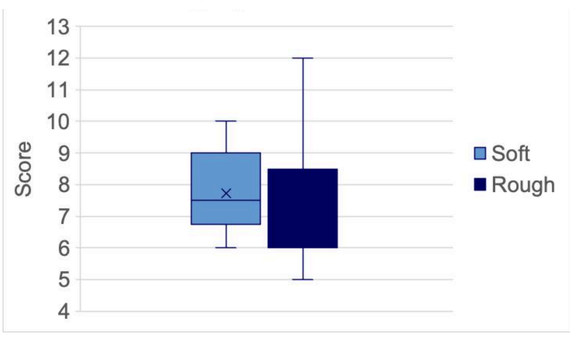
The results from the NASA-TLX survey also had **no statistically significant results** (graph 2). People reported more frustration with the rough vocals, especially during the symbol search task. Interviews also showed that **rough vocals were more noticeable**. This increased noticeability may have contributed to the higher frustration.

Study participants found the **general workload** to be somewhat **higher in the symbol search task while listening to soft music**. This contradicts the task results which show slightly higher scores in both tasks while listening to soft vocals (graph 3). Graph 4 shows very similar means between vocal types, while deviation is almost doubled for tasks performed with the rough vocals.

Graph 3. Symbol Search Task results



Graph 4. Digit Span Task results



Conclusion

In conclusion, no significant differences in participants' cognitive performances were observed between music in rough and soft vocals. The effect of vocal type appears to depend on task type.
Less cognitive damanding tasks might have showed more difference in vocal preference.

References

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